ভারতীয় প্রযুক্তিবিদ্যা প্রতিষ্ঠান খড়্গপুর
भारतीय प्रौद्योगिकी संस्थान खड़गपुर
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Data Systems

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Young Scientist: **Lindau Nobel Laureate Meetings**

Postdoc: **King's College London** & **The Alan Turing Institute**

PhD: **IIT Indore** MTech: **IIITDM Jabalpur**

LINDAU NOBEL LAUREATE MEETINGS

HEIDELBERG LAUREATE FORUM

KING'S College LONDON

The Alan Turing Institute

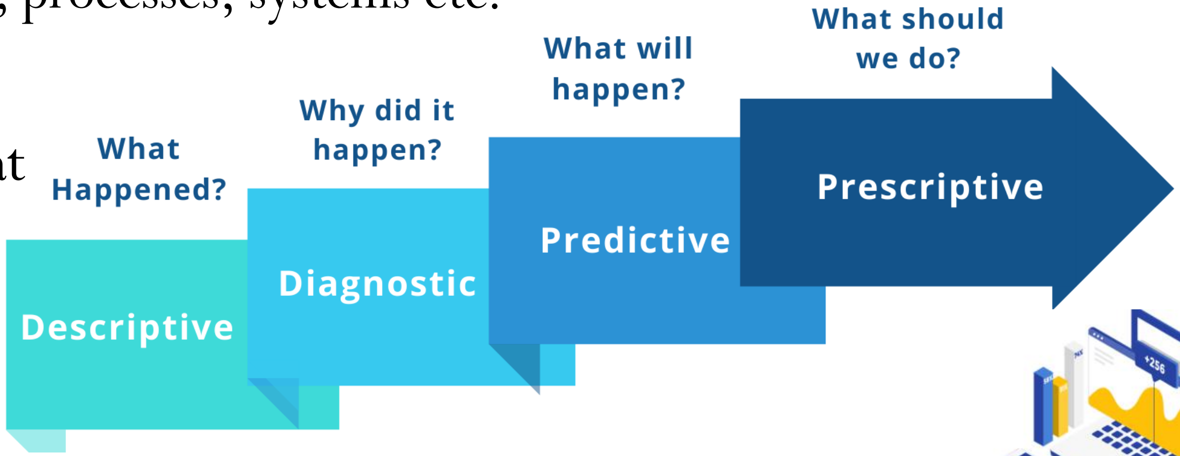
Indian Institute of Technology Indore
भारतीय प्रौद्योगिकी संस्थान इंदौर

PDPM
Indian Institute of Information Technology, Design and Manufacturing, Jabalpur

Data Science, Analytics, KDD, and Big Data

Data Science and Data Analytics

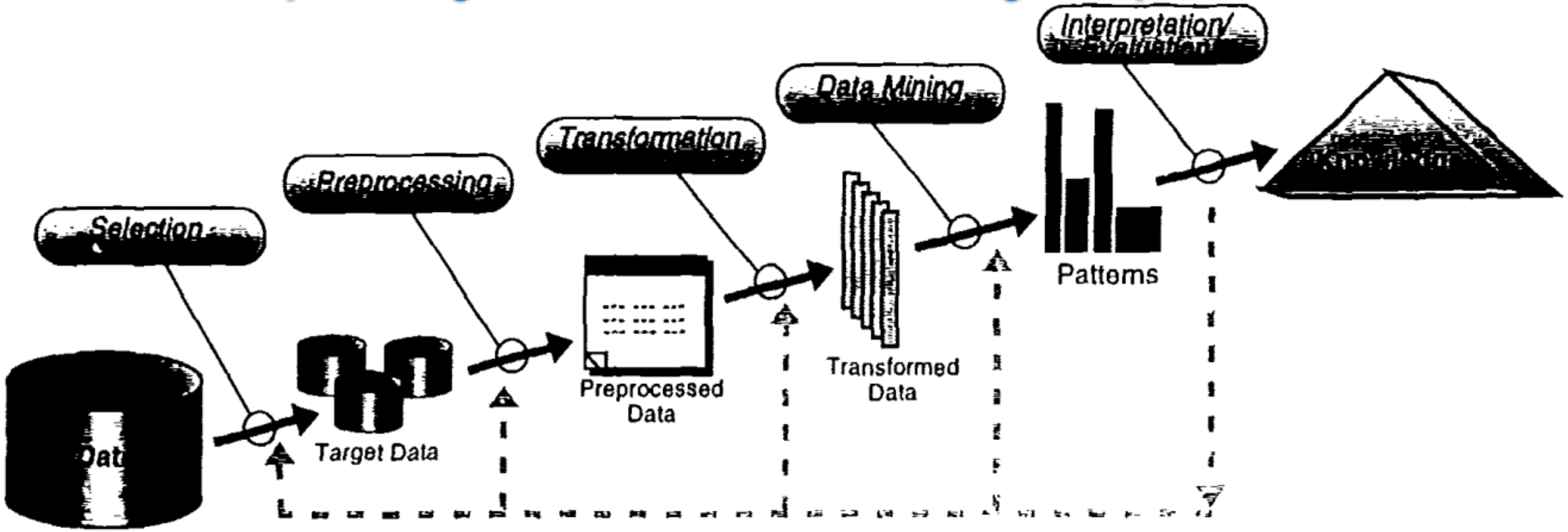
- **Data Science:** interdisciplinary science that
  - deals with data: methods, algorithms, processes, systems etc.
  - Theory oriented
- **Data Analytics:** analysis of data that
  - discovers trends, graph, tables etc.
  - Technology oriented
- Both
  - extracts knowledge and apply actions through structured and unstructured data insights
  - deals with data mining, machine learning, data management, and big data.
- **Data Scientist** and **Data Analyst** applies Data Science and Data Analytics



Knowledge Discovery and Data Mining (KDD)

- "a unifying framework for Knowledge Discovery in Database (KDD)"
- links between data mining, knowledge discovery, and other related field

Selection, Preprocessing, Transformation, Data Mining, Interpretation/Evaluation



Data

Figure 1: An overview of the steps comprising the KDD process.

Fayyad, Usama M., Gregory Piatetsky-Shapiro, and Padhraic Smyth.
"Knowledge Discovery and Data Mining: Towards a Unifying Framework." ***KDD*. 1996.**

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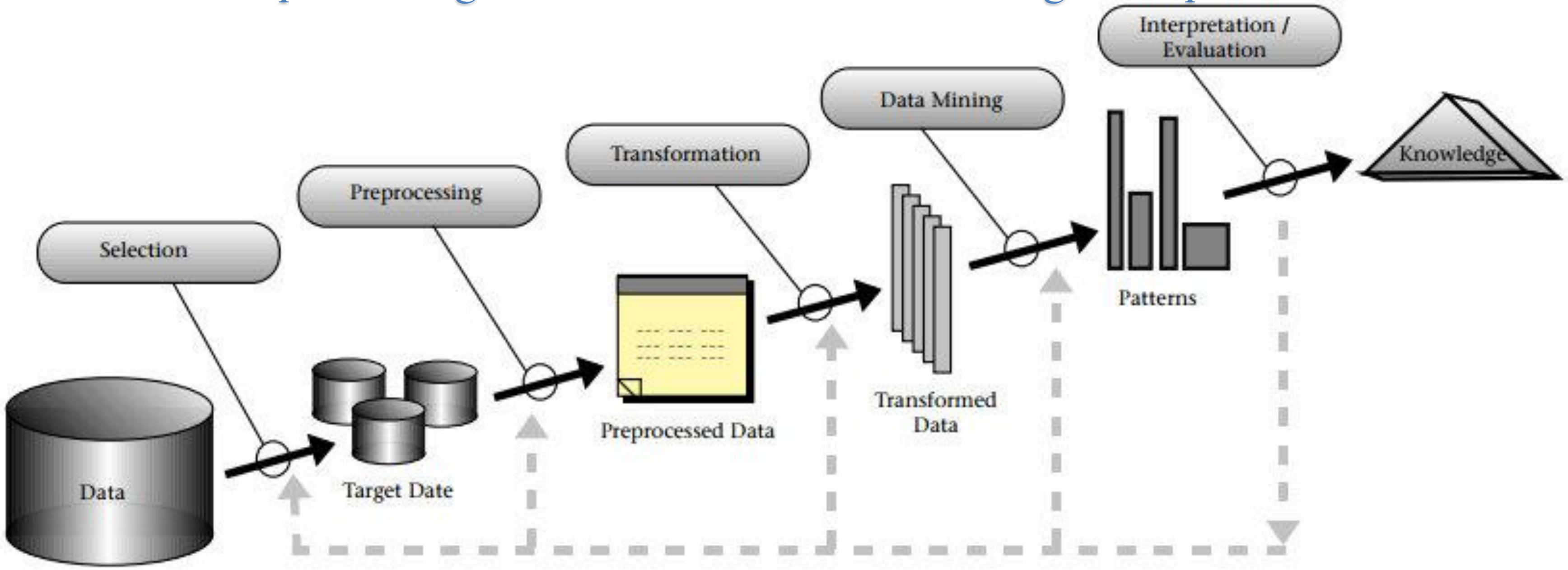


Data

https://www.kdnuggets.com/2016/03/data-science-process-rediscovered.html/2

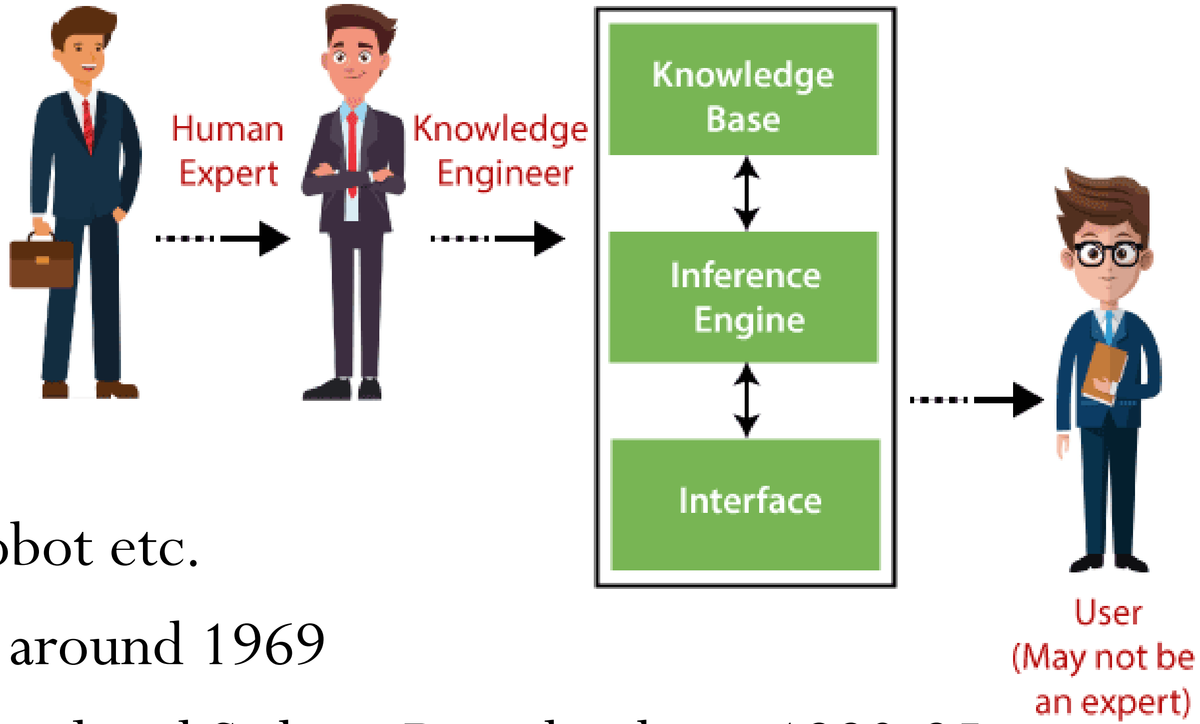
Fayyad, Usama M., Gregory Piatetsky-Shapiro, and Padhraic Smyth. "Knowledge Discovery and Data Mining: Towards a Unifying Framework." ***KDD*.** **1996.**

Big Data

- Big data can be described by the following characteristics:
  - Volume: size large than terabytes and petabytes
  - Variety: type and nature, structured, semi-structured or unstructured
  - Velocity: speed of generation and processing to meet the demands
  - Veracity: the data quality and the data value
  - Value: Useful or not useful
- The main components and ecosystem of Big Data
  - Data Analytics: data mining, machine learning and natural language processing
  - Technologies: Business Intelligence, Cloud computing & Databases
  - Visualization: Charts, Graphs etc.

Expert, System, and Knowledge Engineer

- Expert Systems (1960-74)
  - Explicit rules
- Playing Chess,
- Organic and Biology recommendations
- Solving word problem in Algebra
- Natural Language processing, Mobile Robot etc.
- Backpropagation based Neural Network around 1969
- Association Rule Mining by Rakesh Agrawal and Srikant Ramakrishnan 1993-95
- Big Files and Google File System in 1995-2005 by Larry Page, Sergey Brin and Sanjay Ghemawat, Jeff Dean et al.



Distributed File System

Big Files to Google File System (GFS)

- Earlier Google effort, "BigFiles", developed by Larry Page and Sergey Brin.
  - Supervisors: Hector Garcia-Molina, Rajeev Motwani, Jeff Ullman, and Terry Winograd
- "Big File" was regenerated as "Google File System" by Sanjay Ghemawat, et al.
- Google File System (GFS)
  - "It is widely deployed within Google as the storage platform for the generation and processing of data used by Google service as well as research and development efforts that require large data sets." 2003
  - "The largest cluster to date provides hundreds of terabytes of storage across thousands of disks on over a thousand machines, and it is concurrently accessed by hundreds of clients." 2003

http://infolab.stanford.edu/~backrub/google.html

Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung. "The Google file system." Proceedings of the nineteenth ACM symposium on Operating systems principles. 2003.

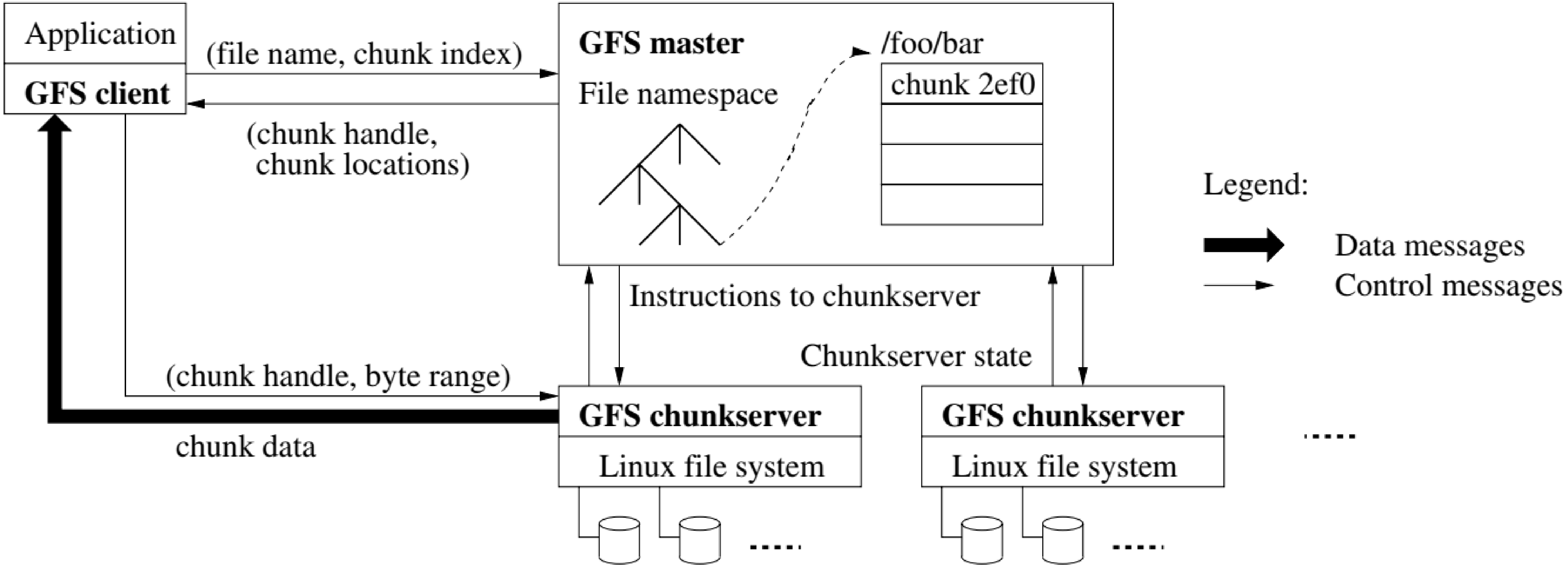
GFS to HDFS

- Google File System (GFS) has similar open-source
  - "Hadoop Distributed File System (HDFS)"
- GFS and HDFS are distributed computing environment to process "Big Data".
- GFS and HDFS are not implemented in the kernel of an operating system, but they are instead provided as a userspace library.
- GFS and HDFS properties
  - Files are divided into fixed-size chunks of 64 megabytes.
  - Scalable distributed file system for large distributed data intensive applications.
  - Provides fault tolerance.
  - High aggregate performance to a large number of clients.

https://en.wikipedia.org/wiki/Google_File_System

Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung. "The Google file system." Proceedings of the nineteenth ACM symposium on Operating systems principles. 2003.

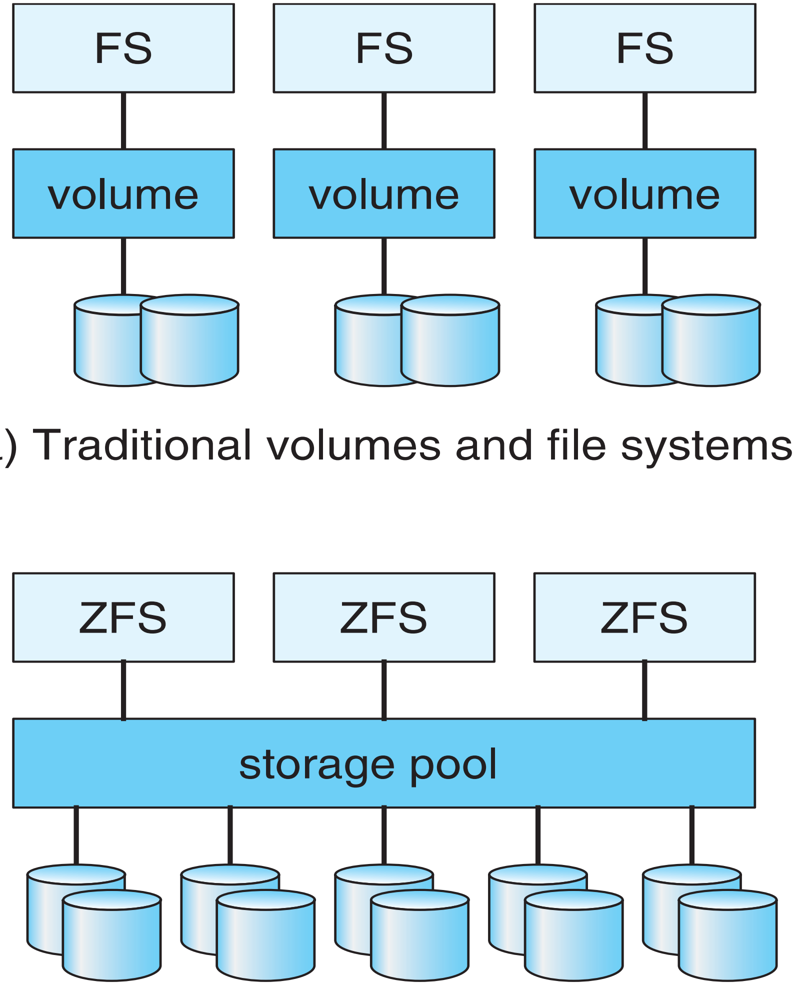
Google File System (GFS)



https://en.wikipedia.org/wiki/Google_File_System
https://sites.google.com/site/gfsassignmentwiki/home
Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung. "The Google file system." Proceedings of the nineteenth ACM symposium on Operating systems principles. 2003.

Traditional and Pooled Storage

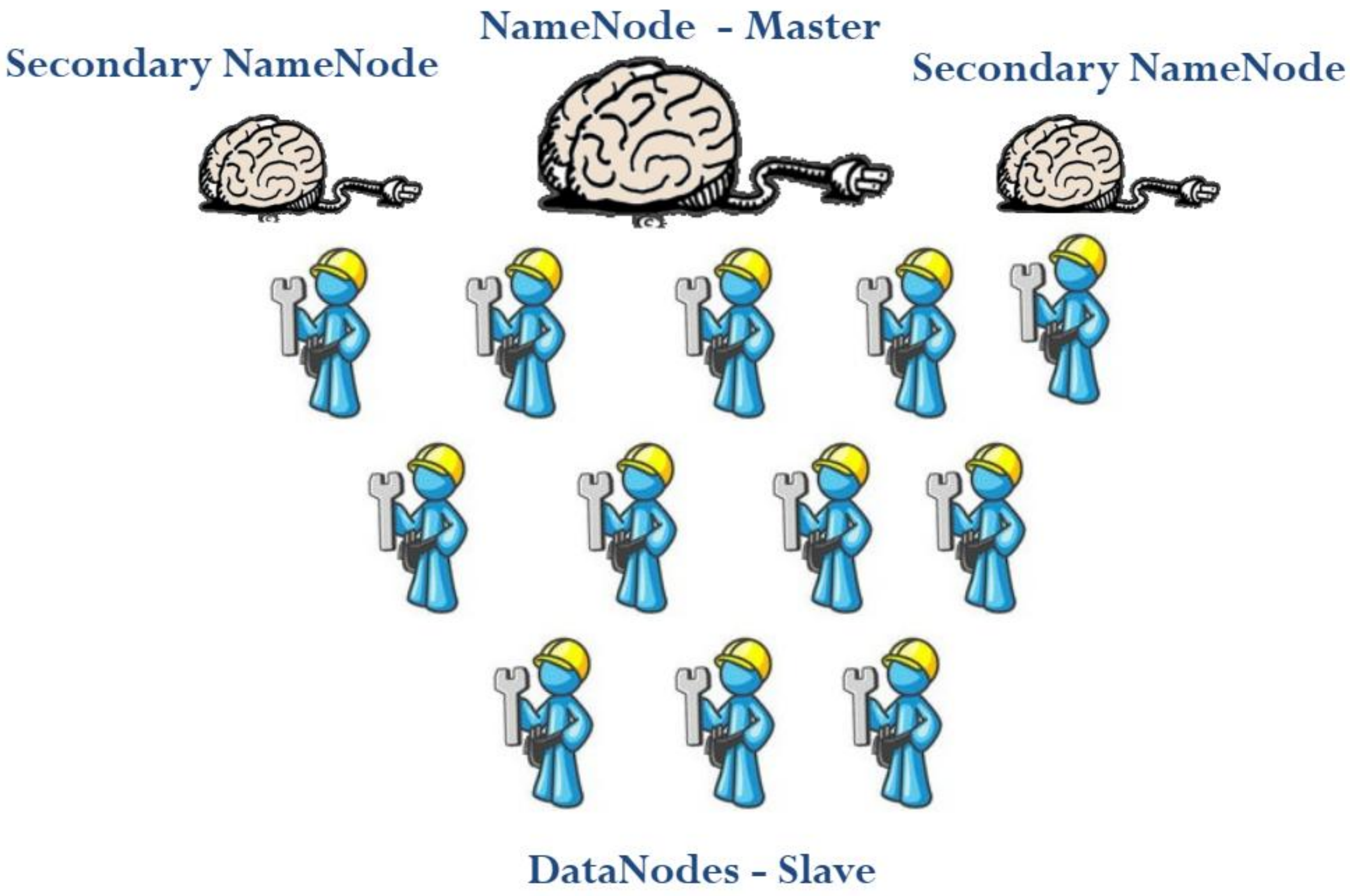
- File System (FS)
- Example: Hadoop Distributed File System (HDFS)
- **Horizontally scalable**



(a) Traditional volumes and file systems.

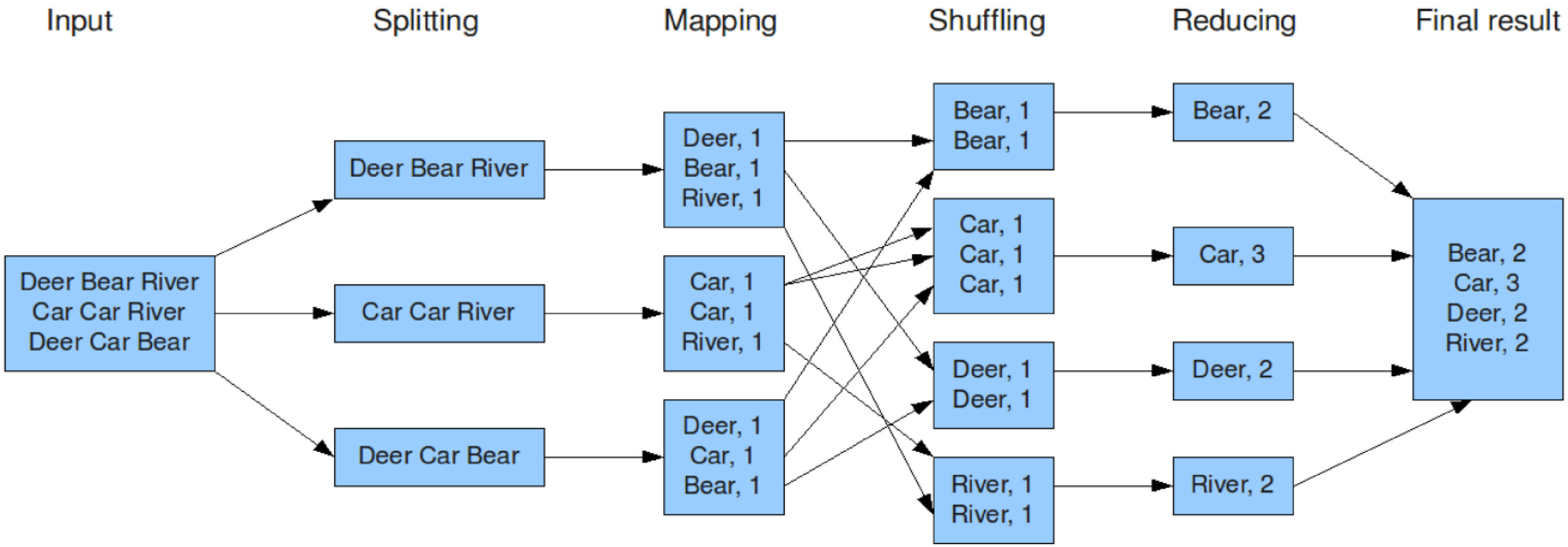
(b) ZFS and pooled storage.

HDFS Force


NameNode - Master
Secondary NameNode
Secondary NameNode
DataNodes - Slave

Word Count

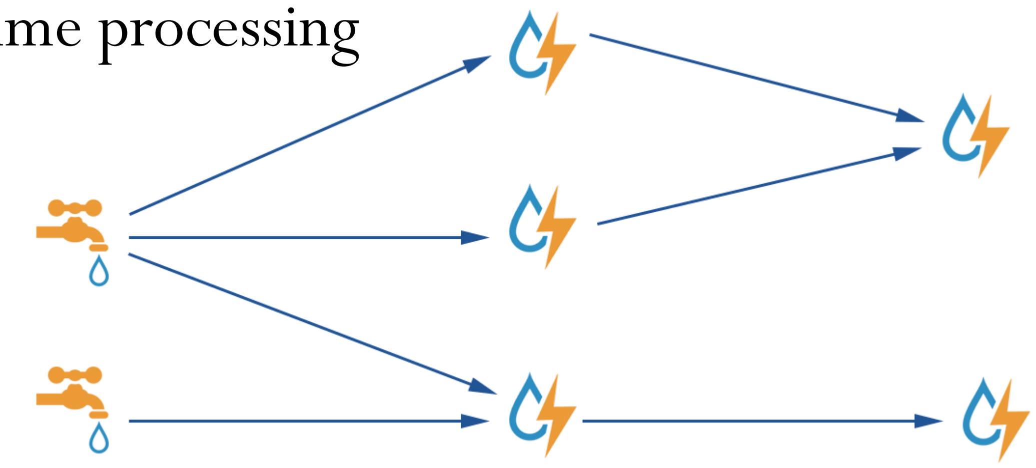
- **Map:** Input lines of text to breaks them into words gives outputs for each word <key = word, value =1 >
- **Reduce:** Input <word, 1> output <word, + value>



Storm

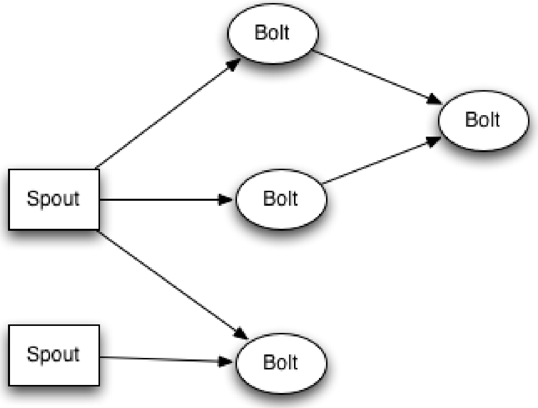
Storm

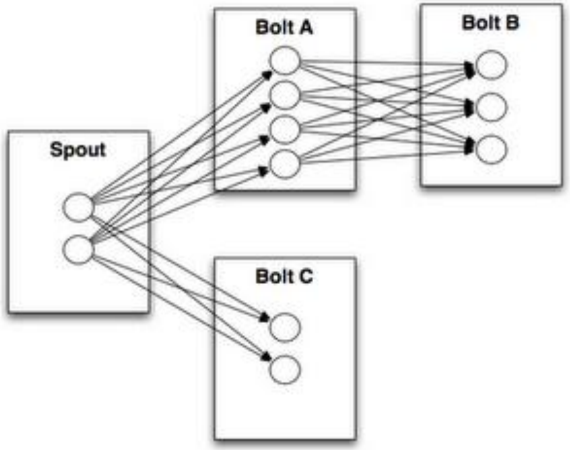
- Reliably for processing unbounded streams of data
- Hadoop for batch processing. Storm real-time processing
- Realtime analytics
- Online machine learning
- Continuous computation
- Distributed RPC.
- A million tupples can be processed per second per node.
- It is scalable, fault-tolerant, guarantees your data will be processed, and is easy to set up and operate.



Storm

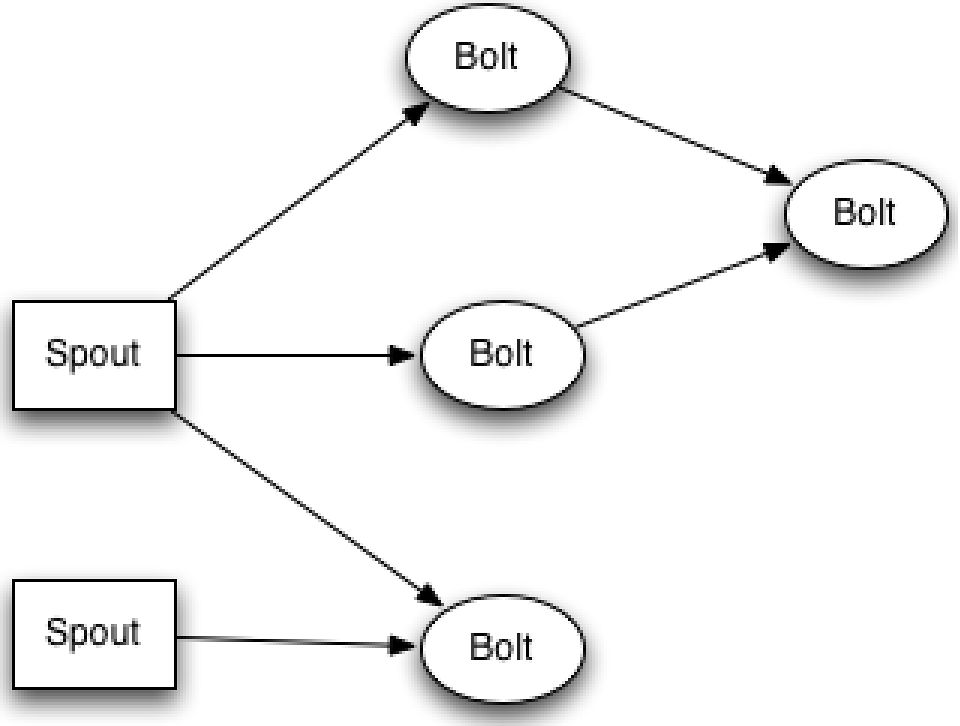
- Apache Storm is a free and open source distributed Realtime computation system.
- Apache Storm makes it easy to reliably process unbounded streams of data, doing for Realtime processing what Hadoop did for batch processing.
- Apache Storm integrates with the queueing and database technologies you already use.
- An Apache Storm topology consumes streams of data and processes those streams in arbitrarily complex ways, repartitioning the streams between each stage of the computation however needed.





Storm

- Topologies: analogous to a MapReduce job. MapReduce job finishes, whereas a topology runs forever or until you kill it.
  - A topology is a graph of spouts and bolts that are connected with stream groupings.
- Streams: is an unbounded sequence of tuples that is processed and created in parallel in a distributed fashion.
  - Streams are defined with a schema that names the fields in the stream's tuples.
  - Schema are integers, longs, shorts, bytes, strings, doubles, floats, booleans, and byte arrays.
  - Every stream is given an id when declared.



Storm@Twitter

- Rectifying incorrect/irrelevant hashtags for tweets
- "Named hashtag recognition" – Aggregate processing of tweets for sentiment and opinion mining
- Topic models to recommend hashtags based on topic distributions
- Incremental learning

List of Feature Scores

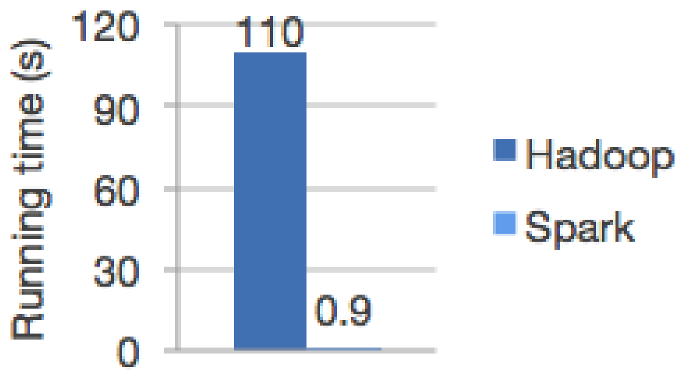
Tweet text	Similarity Score
Temporal Information	Recency Score
Popularity	Social Trend Score
@mentions	Attention score
Favorites	Favorite score
Mutual Friends	Mutual Friend Score
Mutual Followers	Mutual Follower Score
Co-occurrence of hashtags	Common Hashtags Score
Follower-Followee Relation	Reciprocal Score

Spark

Apache Spark

- Unified analytics engine for large-scale data processing.
- Speed: Run workloads 100x faster.
- Both batch and streaming data, using Directed Acyclic Graph (DAG) scheduler, a query optimizer, and a physical execution engine.
- Ease of Use: Write applications quickly in Java, Scala, Python, R, and SQL.
- Spark offers 80+ high-level operators to build parallel apps.

https://spark.apache.org/



Spark: Unified Big Data Analytics

- New applications of Big data workloads on Unified Engine of
  - Streaming, Batch, and Interactive.
- Composability in programming libraries for big data and encourages development of interoperable libraries
- Combining the SQL, machine learning, and streaming libraries in Spark

Zaharia, Matei, et al. "Apache spark: a unified engine for big data processing." *Communications of the ACM* 59.11 (2016): 56-65.

Spark Streaming

- Build scalable fault-tolerant streaming applications.
- Write streaming jobs -- Same Way -- Write batch jobs
  - Counting tweets on a sliding window
- Reuse the same code for batch processing
  - Find words with higher frequency than historic data

Batch processing takes N unit time to process M unit of data

Batch processing **takes N+x unit time** to process **M+y unit of data**

Stream processing takes N unit time to process M unit of data

Stream processing **takes x unit time** to process **M+y unit of data**

https://spark.apache.org/streaming/

Spark GraphX

- Spark's API for graphs and graph-parallel computation
- Fast Speed for graph algorithms
- GraphX graph algorithms
  - PageRank
  - Connected components
  - Label propagation
  - SVD++
  - Strongly connected components
  - Triangle count

https://spark.apache.org/graphx/

Spark MLlib

- Spark's scalable machine learning library
- Spark MLlib algorithms
  - Classification: logistic regression, naive Bayes,…
  - Regression: generalized linear regression, survival regression,…
  - Decision trees, Random forests, and Gradient-boosted trees
  - Recommendation: Alternating Least Squares (ALS)
  - Clustering: K-means, Gaussian mixtures (GMMs),…
  - Topic modeling: Latent Dirichlet Allocation (LDA)
  - **Frequent itemsets, Association rules, and Sequential pattern mining**

https://spark.apache.org/graphx/

Apriori algorithm: Association Rule Mining

- Let I = $\{i_1, i_2, \dots.\ i_m\}$ be a set of literals, called items.
- *Support* of a rule X →Y is the percentage of transactions that contain both X and Y.
- *Confidence* of a rule is percentage the if-then statements (X →Y) are found true
- Find all rules that satisfy a user-specified *minimum support* and *minimum confidence*

TID	Transaction Items
1	Bread, Jelly, PeanutButter
2	Bread, PeanutButter
3	Bread, Milk, PeanutButter
4	Beer, Bread
5	Beer, Milk

{Bread} → {PeanutButter} (Sup = 60%, Conf = 75%)
{PeanutButter} → {Bread} (Sup = 60%, Conf = 100%)
{Beer} → {Bread} (Sup = 20%, Conf = 50%)
{PeanutButter} → {Jelly} (Sup = 20%, Conf = 33.33%)
{Jelly} → {PeanutButter} (Sup = 20%, Conf = 100%)
{Jelly} → {Milk} (Sup = 0%, Conf = 0%)

Rakesh Agrawal, Tomasz Imieliński, and Arun Swami. "Mining association rules between sets of items in large databases." SIG-MOD. 1993.
Ramakrishnan Srikant, and Rakesh Agrawal. "Mining Generalized Association Rules."VLDB 1995.

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- Find all rules that satisfy a user-specified *minimum support* and *minimum confidence*
  - 100% of transactions that purchase *PeanutButter* (antecedent) also purchase *Bread* (consequent). The number 100% is the confidence factor of the rule
    - [PeanutButter] → [Bread] (Sup = 60%, Conf = 100%)
  - 98% of customers who purchase *Tires* and *Auto accessories* also buy some *Automotive services*; here 98% is called the confidence of the rule.
    - [*Auto Accessories*], [*Tires*] → [*Automotive Services*] 98%

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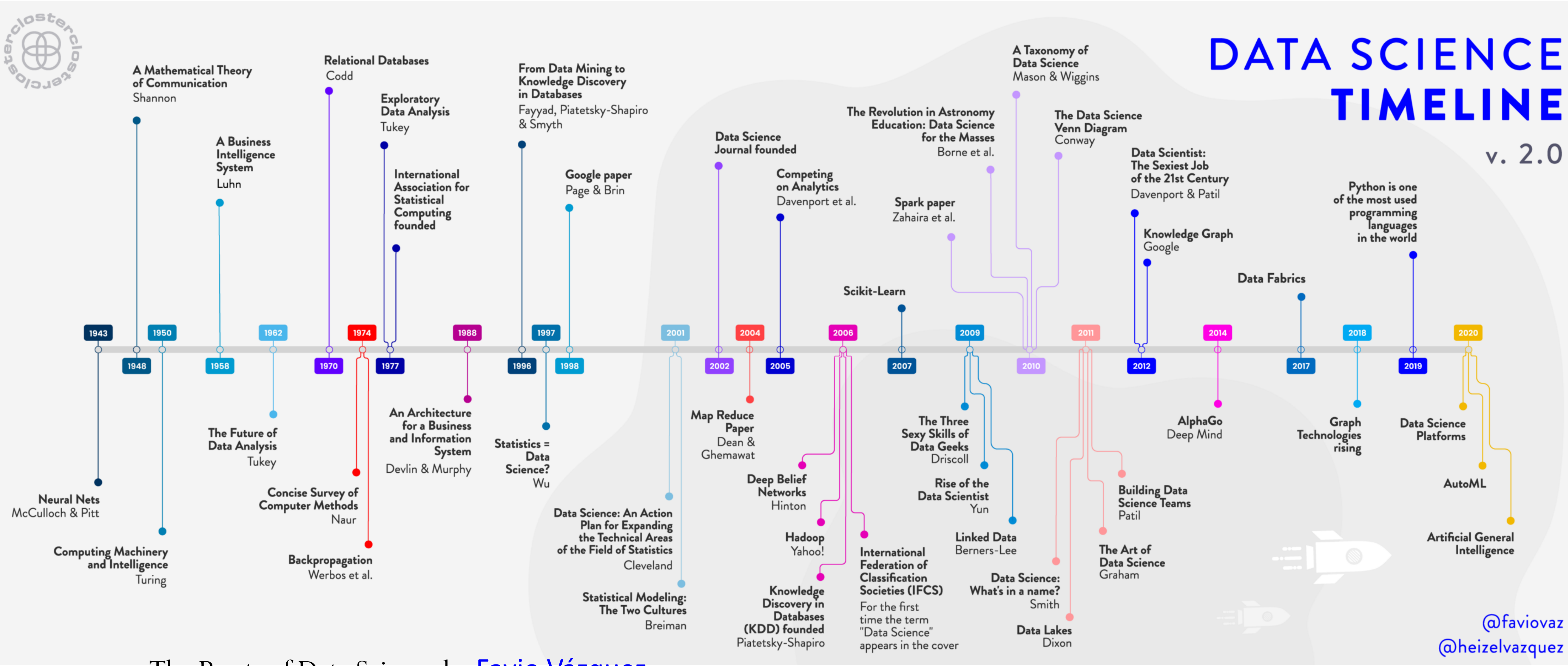
Data Science



The Roots of Data Science by [Favio Vázquez](https://towardsdatascience.com/the-roots-of-data-science-77c71115229)

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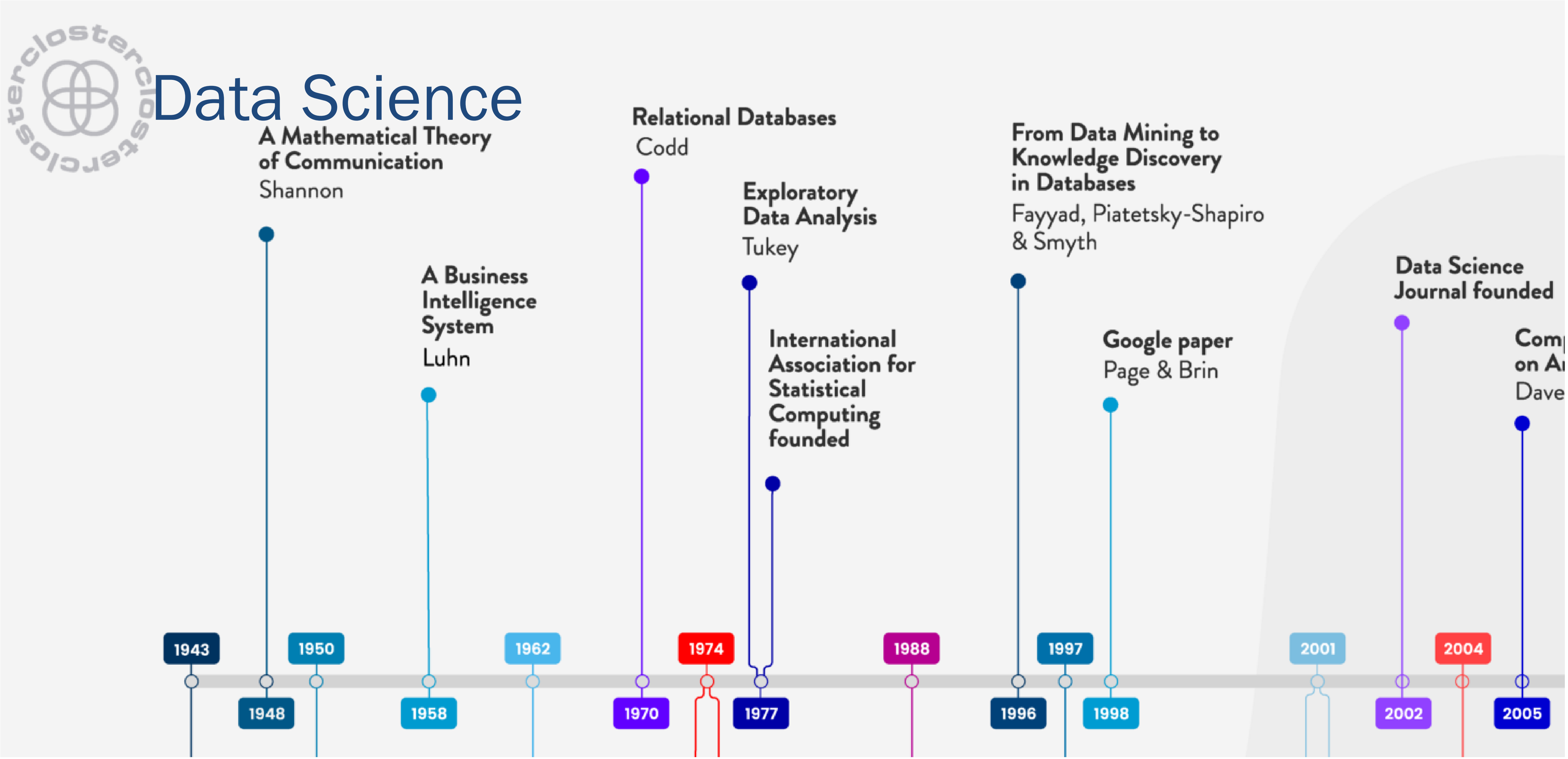

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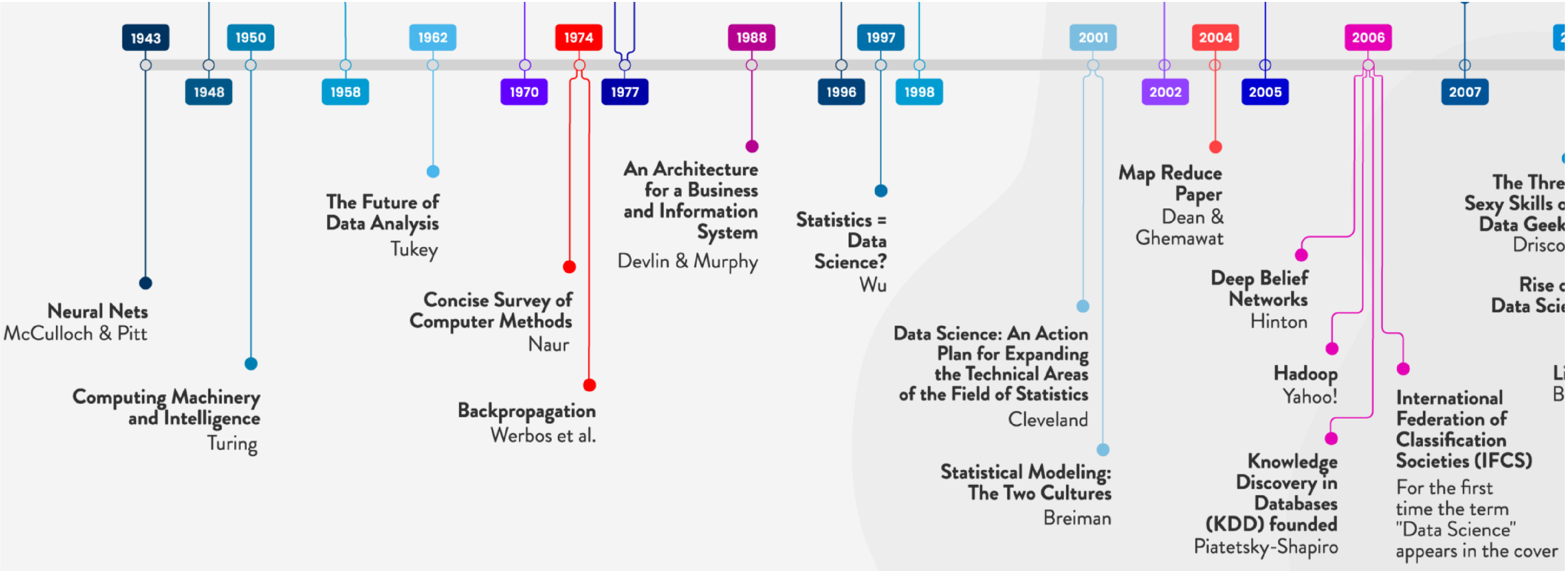

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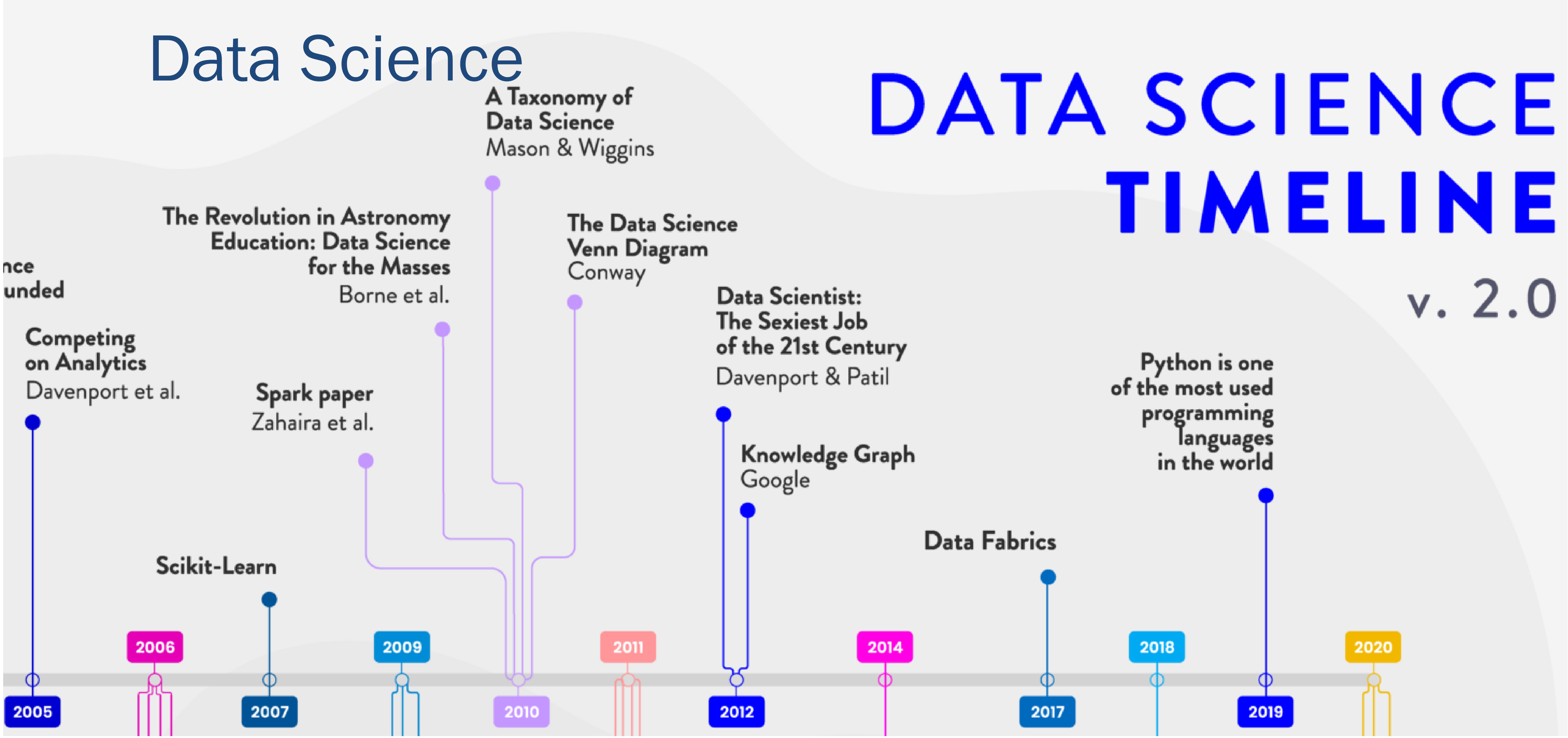

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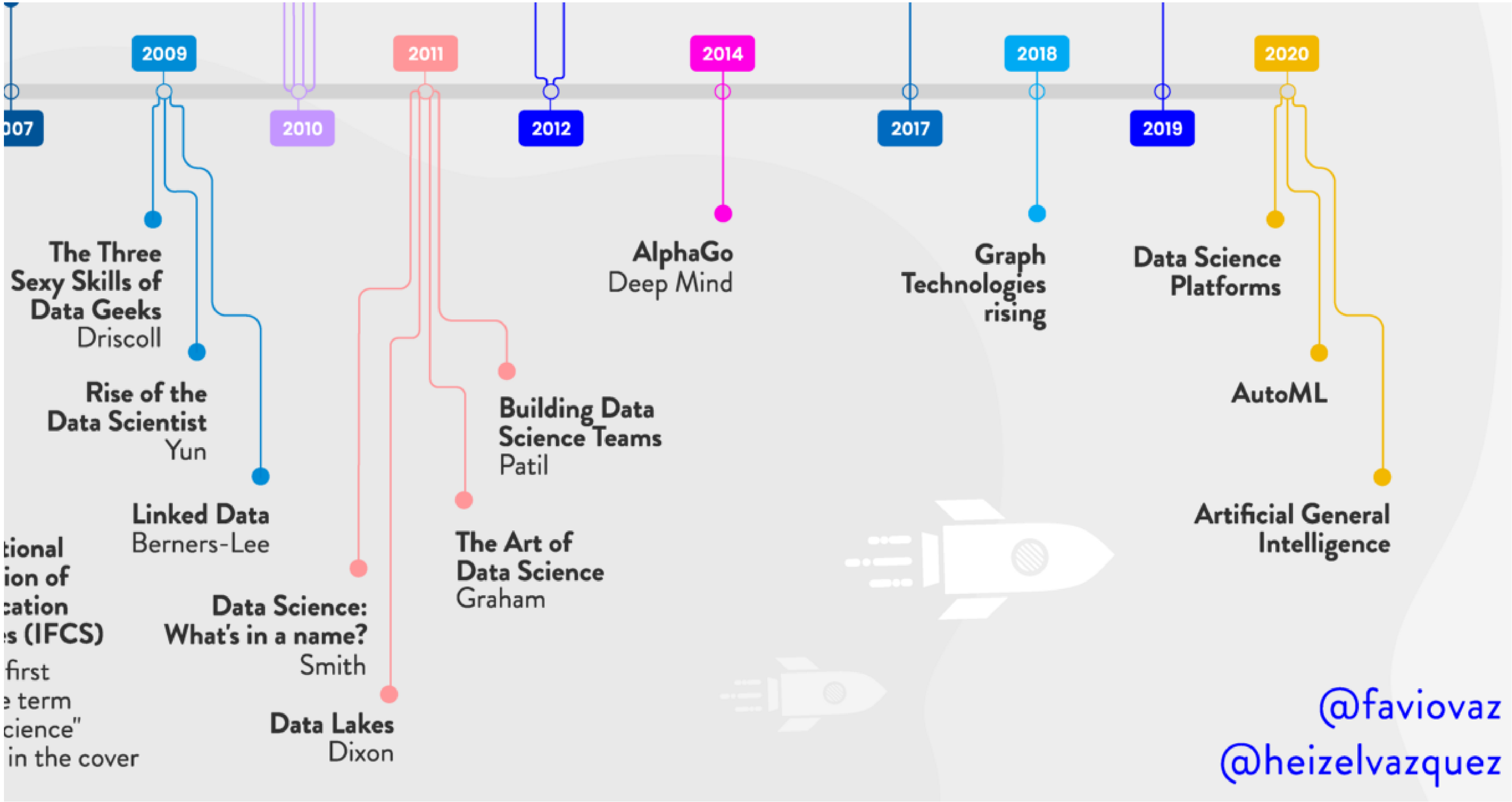

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תודה רבה

Hebrew

Ευχαριστώ

Greek

Danke

German

Спасибо

Russian

धन्यवादः

Sanskrit

Merci

French

ধন্যবাদ

Bangla

நன்றி

Tamil

شكراً

Arabic

ಧನ್ಯವಾದಗಳು

Kannada

Thank You

English

നന്ദി

Malayalam

Grazie

Italian

多謝

Traditional Chinese

ధన్యవాదాలు

Telugu

આભાર

Gujarati

ਧੰਨਵਾਦ

Punjabi

धन्यवाद

Hindi & Marathi

Gracias

Spanish

多谢

Simplified Chinese

https://sites.google.com/site/animeshchaturvedi07

Obrigado

Portuguese

ありがとうございました

Japanese

ขอบคุณ

Thai

감사합니다

Korean